Monday Warm Up: Unit 7: Forces V and Uniform Circular Motion, II

Prof. Jordan C. Hanson

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1 Memory Bank

- $s = r\theta$ ... Relationship between arc length and angular displacement.
- $v = r\omega$ ... Relationship between tangential velocity and angular velocity.
- $\tau = \vec{r} \times \vec{F}$ ... Definition of torque.
- $W = \vec{F} \cdot \Delta\vec{x}$ ... Definition of work (linear).
- $W = \vec{\tau} \cdot \Delta\vec{\theta}$ ... Definition of work (angular).
- $\tau = I\alpha$ ... Magnitude of work, angular acceleration, and moment of inertia, I.
- $I = mr^2$ ... Moment of inertia of a point mass m rotating with radius r.
- **For sytems in static equilibrium, (1) the net force is zero, and (2) the net torque is zero.**

2 Dot Product, and Cross Product

1. Solve:
   - $\vec{a} = 2\hat{i} + 2\hat{j}$, $\vec{b} = -4\hat{i} + 4\hat{j}$. $\vec{a} \cdot \vec{b}$:
   - $\vec{a} = 2\hat{i} + 2\hat{j}$, $\vec{b} = -4\hat{i} + 4\hat{j}$. $\vec{a} \times \vec{b}$:

3 Torque and Stability

1. A steel beam is left balancing on one support in the middle. A 40 kg pile of bricks rests 3 m from the origin, and a 60 kg sack of concrete lies 2 m from the origin. (a) What is the net torque? (b) Suppose the *moment of inertia* is 14×10^3 kg m^2. What is the angular acceleration? (c) What is the angular speed after 2 seconds?
2. Relocate either the bricks or concrete in the previous exercise to achieve balance.
3. The formula for the *moment of inertia* of a point mass m rotating in a circle with radius r is $I = mr^2$. What would it be for two point masses rotating?
4. Suppose we are pulling a rip chord to start a boat motor. We spin an internal gear shaped like a disc, and it has a moment of inertia $\frac{1}{2}MR^2$. (a) If the gear is 1 kg, and the radius is 8 cm, what is the moment of inertia in kg m^2? (b) What torque is required from us to spin the gear with $\alpha = 2$ rad/s^2?